

Assignment 7

Due Friday 10 April at the beginning of class

This Assignment is based on the slides, especially [weeks 10 & 11](#) and [week 12](#).

DO THE FOLLOWING PROBLEMS.

P17. First see the “main idea” about the definition of $C_c^\infty(\Omega)$ in the weeks 10 & 11 slides.

(a) Suppose $\Omega \subset \mathbb{R}^d$ is open. Suppose $g \in L_{loc}^1(\Omega)$ and $\varphi \in C_c^\infty(\Omega)$. Prove that $gD^\alpha\varphi \in L^1(\Omega)$, thus that $\int_\Omega gD^\alpha\varphi \, dm$ is well-defined.

(b) Show that $C(\Omega) \subset L_{loc}^1(\Omega)$. Then, for each dimension d , construct an example of Ω and $f \in C(\Omega)$ showing that “ $C(\Omega) \subset L^1(\Omega)$ ” is false.

P18. Here is an important case using the weak derivative concept in 1D.

Suppose $\Omega = (a, b)$ is a bounded open interval, and let $a = t_0 < t_1 < \cdots < t_n = b$ be a mesh on a closed interval $[a, b]$. As in the slides, the P_1 finite element space is defined as the set of functions which are continuous on $[a, b]$, and which are piecewise linear on each cell (subinterval) $[t_{k-1}, t_k]$. Show that $P_1 \subset H^1(\Omega)$.

P19. Here is another case using the weak derivative concept in 1D. Note the key facts below that s and s' are continuous on $[a, b]$. However, generally s'' is not continuous, or even defined at all points in $[a, b]$.

Suppose $a = t_0 < t_1 < \cdots < t_n = b$ is a mesh on a closed interval $[a, b] \subset \mathbb{R}^1$, and consider values $y_0, y_1, \dots, y_n \in \mathbb{R}^1$. An interpolating *cubic spline* is a function $s \in C^1([a, b])$ such that $s(x)$ on each subinterval is a cubic polynomial,

$$s(x) = a_k + b_k(x - x_{k-1}) + c_k(x - x_{k-1})^2 + d_k(x - x_{k-1})^3 \quad \text{on } [t_{k-1}, t_k],$$

for $k = 1, \dots, n$, and so that $s(t_k) = y_k$, for $k = 0, 1, \dots, n$. It is a fact that for the data (t_k, y_k) there is a unique function $s(x)$ with these properties. (You do not need to prove this.) Give a formula for the weak second derivative s'' , on $[a, b]$, and show that it satisfies the definition of a weak second derivative. Approximately sketch the spline $s(x)$ and its second derivative $s''(x)$ in a simple case, for instance with $n = 3$.

P20. Recall that the norm in $H^k(-\pi, \pi)$ satisfies $(\|f\|_{H^k})^2 = \sum_{\ell=0}^k \int_{-\pi}^{\pi} |f^{(\ell)}(x)|^2 dx$. Here $f^{(\ell)}$ denotes the weak derivative of order ℓ .

(a) Suppose $f \in H^1(-\pi, \pi)$. Using the formulas for Fourier series,¹ prove the derivative formula

$$f'(x) = \frac{1}{\sqrt{2\pi}} \sum_{n \in \mathbb{Z}} in c_n e^{inx},$$

¹See “recall Fourier series in $L^2(-\pi, \pi)$ ” in the [week 12](#) slides.

with convergence in L^2 , where c_n are the Fourier coefficients of f .

(b) Show that

$$\|f\|_{H^k}^2 = \sum_{n \in \mathbb{Z}} \left(\sum_{j=0}^k n^{2j} \right) |c_n|^2.$$

(There is no need to justify derivative formulas, beyond what you did in part **(a)**.)

P21. This example asks you to do the details of an example shown in the slides. It shows that H^1 contains discontinuous functions for planar domains. To say it more precisely, there are equivalence classes in H^1 which have no continuous representative.

Let $D = \{x \in \mathbb{R}^2 : |x|^2 < 1/4\}$ be an open disk, centered at the origin, with radius $1/2$. Let $g(x) = \ln(r)$ and $p(x) = \ln(|\ln(r)|)$, in polar coordinates where $|x| = r$, defined a.e. on D . (These functions are not defined at the origin.) Exactly compute, by hand, the L^2 and H^1 norms of g and p . Then use a computer to plot g and p over $0 < r < 1/2$.

P22. For $z = x + iy$ a complex number, define $p(z) = z^3 - 5z^2 + 2z - 1$. Show that the real and imaginary parts of p , regarded as real-valued functions of (x, y) on $\mathbb{R}^2$, are harmonic.